CLIENT	**:**	**Raviraj Reality**
SITE	**:**	**Raviraj Reality , Pune**
LOCATION	**:**	**PUNE, MHARASHTRA**
NO.	**:**	**-**
DOCUMENT NO.	**:**	**DBR-R00**

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Rev No.	**Date**	**Description**	**Prepared By**	**Checked By**	**Approved By**

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1. Design Philosophy

This document pertains to structural design of **Retractable roof at Raviraj Reality**, Pune, Maharashtra. The overall size of the frame is 11.1 x 11.3 x 2.791m high. The main sliding door frame is of aluminium 6036-T6 material. The wind load is worked out from IS 875 Part III. The various DL and LL acting at different points on frame are transferred to the nearest columns through the grid of beams.

The complete frame is modelled as 3D frame and analysed/designed for worst combination of loads through STAAD.pro. Input and output files are enclosed which are self-explanatory. From the STAAD output, the support reactions for various primary load cases are tabulated and the base plate, shear keys and foundation bolts are designed accordingly. Foundation design shall be done using appropriate load combinations for reactions given in this document.

2. DESIGN CODES AND STANDARDS

Code / Standard	Applicability	Remark
IS800-2007	Design of steel structures	--
IS 875-Part III	For wind load calculation	--

3. MATERIAL SPECIFICATION FOR CONSTRUCTION

Sr.no	Material	Specification	Minimum Strength
1	Aluminium	Aluminium 6063-T6	Fy = 416 N/mm^2

4. LOAD CALCULATIONS

4.1 DEAD LOAD

SELFWEIGHT FACTOR = 1

Dead load for glass/sheet assumed = 5 Kg/m^2 i.e. 0.05 kN/m^2

4.2 LIVE LOAD

Live load assumed = 0.75 kN/m^2

4.3 WIND LOAD

For wind load calculation, following IS code is referred: - "IS 875 Part III"

DESIGN WIND PRESSURE CALCULATION				
Sr no:	**Description**	**Value**	**Unit**	**Remarks**
1	V_b Basic Wind Speed	39	m/s	As per Apendix A (IS-875 Part 3)
2	K1 (risk coefficient)	1		For Design life span of 50 yrs. As per IS 875 Part III (table-1)
3	K2 (Terrain roughness and height factor)	1		As per IS 875 Part III (table-2), Cat-2, Ht 33M
4	K3 (topography factor)	1		Clause 6.3.3 IS 875 part III
5	K4 (Importance factor for Cyclonic region.)	1		Clause 6.3.4 IS 875 part III
6	K_a (Area Averaging factor)	0.80		As per IS 875 Part III (table-4)
7	K_d (Wind Directionality Factor)	0.9		Clause 7.2.1 IS 875 part III
8	K_c (Combination Factor)	0.9		Clause 7.3.3.13 IS 875 part III
	Vz=	Vb*K1*K2*K3*K4		
	Vz	**39**	**m/s**	
	Pz =	0.91	kN/m2	Pz =0.6(Vz)^2
	Pd1 =	0.59	kN/m2	Pd1=Pz*Ka*Kd*Kc
	or Pd2 =	0.64	kN/m2	Pd2=0.7*Pz
	Hence Use pd =	0.64	kN/m2	Max of Pd1 and Pd2
	Coefficeint for Pressure Case=	0.6		
	Coefficeint for Suction Case=	0.3		
	Span =	1		
	UDL for Pressure case =	0.4	kN/m	
	UDL for Suction case =	0.2	kN/m	

5. Deflection check at building top in Z direction

Refer node no. –52 of staad file for all load cases

Max. Deflection in Z dir. = 12.8 mm < H/250 = 11100/250 = 44.4mm (IS 800-2007 table 6)

.... Hence ok.

6. PRIMARY LOADS & LOAD COMBINATIONS

6.1 PRIMARY LOADS

Load 1 Dead Load (DL)

Load 2 Live Load (LL)

Load 3 Wind Load Pressure (WLP)

Load 4 Wind Load Suction (WLS)

6.2 LOAD COMBINATIONS

Load Combination for strength:-

1.5 DL + 1.5 LL
1.5 DL + 1.5 WLP
1.5 DL + 1.5 WLS

Load Combination for serviceability:-

DL + LL
DL + WLP
DL + WLS
DL + 0.8 WLP
DL + 0.8 WLS